

Modular Inverses

1 Definition

Modular Inverse

An integer x is a **multiplicative inverse** of a modulo n if

$$ax \equiv 1 \pmod{n}.$$

We write

$$x \equiv a^{-1} \pmod{n}.$$

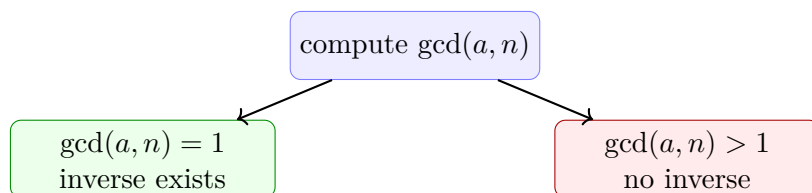
The inverse is the modular version of division. Instead of dividing by a , we multiply by a^{-1} .

2 When Does an Inverse Exist?

Existence Test

$$a^{-1} \pmod{n} \text{ exists if and only if } \gcd(a, n) = 1.$$

If a and n share a factor greater than 1, then every product ax shares that obstruction. It cannot become congruent to 1 modulo n .



3 Connection with Extended Euclid

If

$$\gcd(a, n) = 1,$$

then Bézout's identity gives integers x and y such that

$$ax + ny = 1.$$

Reduce both sides modulo n :

$$ax + ny \equiv 1 \pmod{n}.$$

Since $ny \equiv 0 \pmod{n}$,

$$ax \equiv 1 \pmod{n}.$$

Therefore

$$x \equiv a^{-1} \pmod{n}.$$

Key Point

To find $a^{-1} \pmod{n}$, run Extended Euclid on (a, n) . The coefficient of a is the inverse, reduced modulo n .

4 Algorithm

Modular Inverse by Extended Euclid

```
ModInverse(a, n):  
    (d, x, y) = ExtendedEuclid(a, n)  
  
    if d != 1:  
        return "no inverse"  
  
    return x mod n
```

The final line $x \bmod n$ is needed because Extended Euclid may return a negative coefficient.

5 Example: No Inverse

Problem

Find $2^{-1} \pmod{14}$.

Solution

Check the gcd:

$$\gcd(2, 14) = 2.$$

Since

$$2 \neq 1,$$

the inverse does not exist.

Answer

$2^{-1} \pmod{14}$ does not exist.

6 Example: Direct Reduction

Problem

Find $4^{-1} \pmod{7}$.

Solution

We need x such that

$$4x \equiv 1 \pmod{7}.$$

Try small residues:

$$4 \cdot 1 \equiv 4 \pmod{7},$$

$$4 \cdot 2 \equiv 8 \equiv 1 \pmod{7}.$$

Therefore

$$\boxed{4^{-1} \equiv 2 \pmod{7}.$$

Check

$$4 \cdot 2 = 8 \equiv 1 \pmod{7}.$$

7 Example: Extended Euclid

Problem

Find $7^{-1} \pmod{26}$.

Solution

Run Euclid:

$$26 = 7(3) + 5,$$

$$7 = 5(1) + 2,$$

$$5 = 2(2) + 1.$$

Now back-substitute:

$$1 = 5 - 2(2).$$

Since

$$2 = 7 - 5,$$

we get

$$1 = 5 - 2(7 - 5) = 3(5) - 2(7).$$

Since

$$5 = 26 - 7(3),$$

we get

$$1 = 3(26 - 7(3)) - 2(7).$$

So

$$1 = 3(26) - 11(7).$$

The coefficient of 7 is -11 . Therefore

$$7^{-1} \equiv -11 \pmod{26}.$$

Reduce:

$$-11 \equiv 15 \pmod{26}.$$

Answer

$$7^{-1} \equiv 15 \pmod{26}.$$

8 Example: Negative Coefficient

Problem

Find $5^{-1} \pmod{11}$.

Solution

Use Euclid:

$$11 = 5(2) + 1.$$

Back-substitute:

$$1 = 11 - 5(2).$$

Thus

$$1 = 5(-2) + 11(1).$$

The coefficient of 5 is -2 . Therefore

$$5^{-1} \equiv -2 \pmod{11}.$$

Reduce the negative coefficient:

$$-2 \equiv 9 \pmod{11}.$$

So

$$5^{-1} \equiv 9 \pmod{11}.$$

Always Check the Bézout Identity

The coefficient method works only when the equation is actually

$$1 = ax + ny.$$

If the right side is not 1, the coefficient is not an inverse.

9 Correct Negative Coefficient Example

Problem

Find $3^{-1} \pmod{35}$.

Solution

Use Extended Euclid:

$$35 = 3(11) + 2,$$

$$3 = 2(1) + 1.$$

Back-substitute:

$$1 = 3 - 2.$$

Since

$$2 = 35 - 3(11),$$

we get

$$1 = 3 - (35 - 3(11)).$$

Thus

$$1 = 12(3) - 35.$$

The coefficient of 3 is 12, so

$$\boxed{3^{-1} \equiv 12 \pmod{35}.$$

10 More Quick Examples

Example 1: $5x \equiv 1 \pmod{14}$

We need the inverse of 5 modulo 14.

$$\gcd(5, 14) = 1,$$

so an inverse exists.

Check small multiples:

$$5 \cdot 3 = 15 \equiv 1 \pmod{14}.$$

Therefore

$$\boxed{x \equiv 3 \pmod{14}.$$

Example 2: $35x \equiv 1 \pmod{8}$

First reduce 35 modulo 8:

$$35 \equiv 3 \pmod{8}.$$

So the equation becomes

$$3x \equiv 1 \pmod{8}.$$

Now check:

$$3 \cdot 3 = 9 \equiv 1 \pmod{8}.$$

Therefore

$$\boxed{x \equiv 3 \pmod{8}.$$

Example 3: $40x \equiv 1 \pmod{7}$

First reduce 40 modulo 7:

$$40 \equiv 5 \pmod{7}.$$

So the equation becomes

$$5x \equiv 1 \pmod{7}.$$

Now check:

$$5 \cdot 3 = 15 \equiv 1 \pmod{7}.$$

Therefore

$$x \equiv 3 \pmod{7}.$$

Example 4: $20x \equiv 1 \pmod{3}$

First reduce 20 modulo 3:

$$20 \equiv 2 \pmod{3}.$$

So the equation becomes

$$2x \equiv 1 \pmod{3}.$$

Now check:

$$2 \cdot 2 = 4 \equiv 1 \pmod{3}.$$

Therefore

$$x \equiv 2 \pmod{3}.$$

Fast Pattern

For equations of the form

$$ax \equiv 1 \pmod{n},$$

first reduce a modulo n , then find the number that multiplies it to give remainder 1.

11 Tabular Extended Euclidean Method

A common table for modular inverses tracks quotients and coefficients while running Euclid.

Step	Remainder	Quotient	Coefficient of a
0	n	—	0
1	a	—	1
2	r_2	q_1	s_2
3	r_3	q_2	s_3

When the final nonzero remainder is 1, the matching coefficient of a is the inverse modulo n .

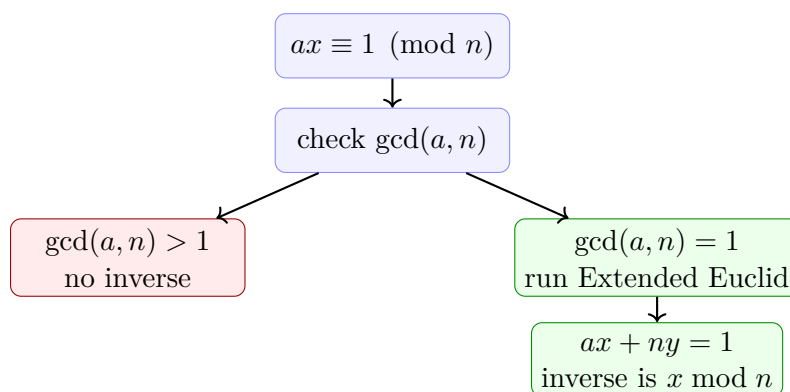
Worked Table: $3^{-1} \pmod{35}$

Step	Remainder	Equation	Coefficient of 3
0	35	35	0
1	3	3	1
2	2	$35 - 11(3)$	-11
3	1	$3 - 1(2)$	12

The coefficient of 3 when the remainder becomes 1 is 12. Hence

$$3^{-1} \equiv 12 \pmod{35}.$$

12 Visual Summary



13 Review Box

Remember

$$\begin{aligned}x &= a^{-1} \pmod{n} \iff ax \equiv 1 \pmod{n}, \\a^{-1} \pmod{n} \text{ exists} &\iff \gcd(a, n) = 1, \\ax + ny = 1 &\Rightarrow x \equiv a^{-1} \pmod{n}.\end{aligned}$$